

- 6 (a) The photoelectric effect is only observed when the frequency of the electromagnetic radiation incident on a particular metal is at least a minimum frequency. It does not depend on the intensity of the radiation. This is the threshold frequency and each metal has its own threshold frequency.

This can only be explained if electromagnetic radiation is made up of discrete quanta of energy known as photons.

The energy of each photon is hf , where h is the Planck constant and f is the frequency of the radiation.

An electron in the metal requires a minimum energy to leave the surface of the metal in the photoelectric effect. This minimum energy is the work function Φ of the metal.

The electron absorbs energy only from one photon in the particulate theory. Hence for photoemission to occur, the energy of each photon $hf \geq \Phi$. The minimum frequency $f = \Phi/h$ is the threshold frequency.

- (b) (i) Solution 1

horizontal-intercept gives the threshold wavelength $\lambda_0 = 4.4 \times 10^{-7} \text{ m}$ as $E_{\text{max}} = 0$

$$\begin{aligned}\Phi &= \frac{hc}{\lambda_0} \\ &= \frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{4.4 \times 10^{-7}} \\ &= 4.5205 \times 10^{-19} = 4.52 \times 10^{-19} \text{ J}\end{aligned}$$

Solution 2

$$\begin{aligned}\frac{hc}{\lambda} &= \Phi + E_{\text{max}} \\ E_{\text{max}}\lambda &= (-\Phi)\lambda + hc\end{aligned}$$

for the graph of $E_{\text{max}}\lambda$ against λ , gradient = $-\Phi$, vertical-intercept = hc

$$\begin{aligned}\text{gradient} &= \frac{(1.10 - 0.10) \times 10^{-25}}{(2.0 - 4.2) \times 10^{-7}} = -4.5455 \times 10^{-19} \\ \Phi &= -\text{gradient} = 4.5455 \times 10^{-19} = 4.55 \times 10^{-19} \text{ J}\end{aligned}$$

- (ii) 1. for $\lambda = 2.0 \times 10^{-7} \text{ m}$, $E_{\text{max}}\lambda = 1.10 \times 10^{-25} \text{ J m}$

$$\begin{aligned}E_{\text{max}} &= \frac{1.10 \times 10^{-25}}{2.0 \times 10^{-7}} = 5.5 \times 10^{-19} \text{ J} \\ eV_s &= E_{\text{max}}\end{aligned}$$

$$\begin{aligned}V_s &= \frac{1.10 \times 10^{-25}}{(2.0 \times 10^{-7})(1.60 \times 10^{-19})} \\ &= 3.4375 = 3.44 \text{ V}\end{aligned}$$

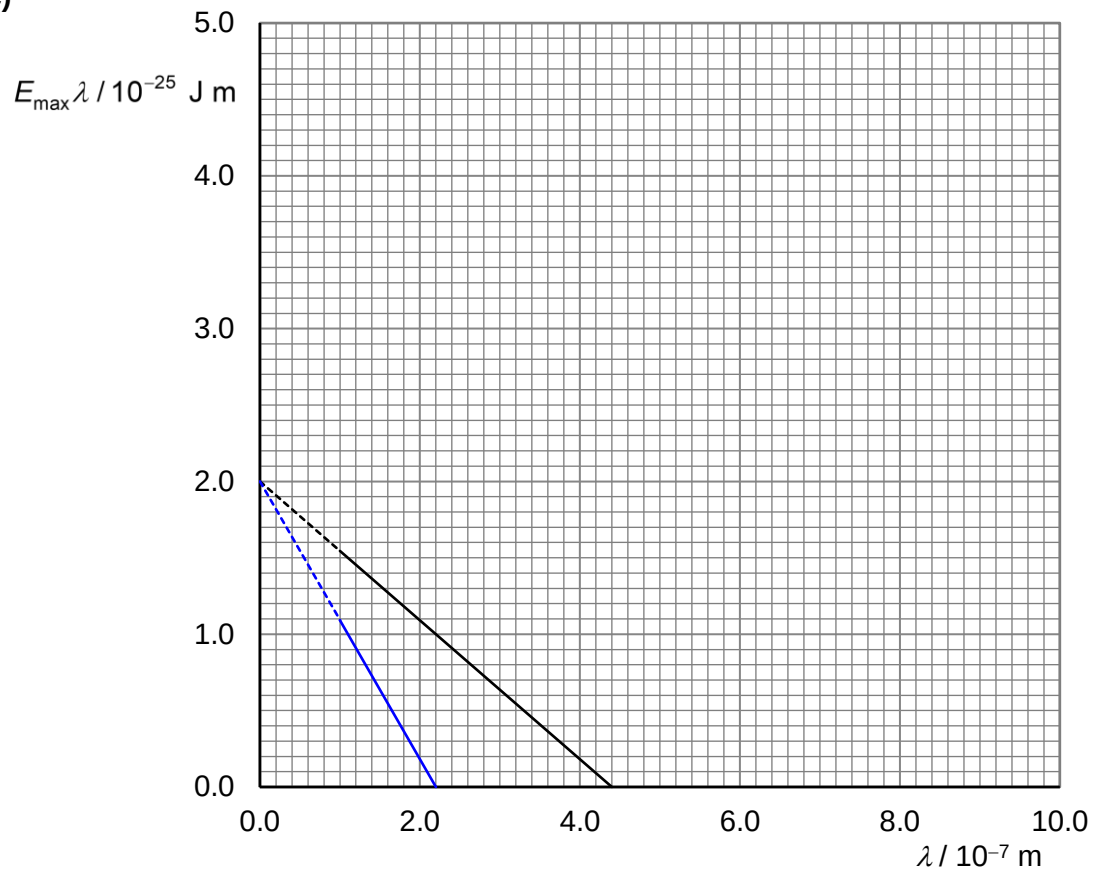
2.

$$\begin{aligned}E &= \frac{1}{2}mv^2 = \frac{1}{2} \frac{(mv)^2}{m} = \frac{p^2}{2m} \\ p &= \sqrt{2mE}\end{aligned}$$

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$$\begin{aligned}\lambda &= \frac{h}{p} \\ &= \frac{h}{\sqrt{2mE_{\max}}} \\ &= \frac{6.63 \times 10^{-34}}{\sqrt{2(9.11 \times 10^{-31})(5.5 \times 10^{-19})}} \\ &= 6.6230 \times 10^{-10} = 6.62 \times 10^{-10} \text{ m}\end{aligned}$$

(c)



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Paper 3 – Section B

7. (a) (i) For a fixed tension (or load) in the string, the speed of the transverse progressive